

Acidity, basicity, and pK_a

Connections

➡ Building on

- Conjugation and molecular stability [ch7](#)
- Curly arrows represent delocalization and mechanisms [ch5](#)
- How orbitals overlap to form conjugated systems [ch4](#)

Arriving at

- Why some molecules are acidic and others basic
- Why some acids are strong and others weak
- Why some bases are strong and others weak
- Estimating acidity and basicity using pH and pK_a
- Structure and equilibria in proton transfer reactions
- Which protons in more complex molecules are more acidic
- Which lone pairs in more complex molecules are more basic
- Quantitative acid/base ideas affecting reactions and solubility
- Effects of quantitative acid/base ideas on medicine design

➡ Looking forward to

- Acid and base catalysis in carbonyl reactions [ch10 & ch11](#)
- The role of catalysts in organic mechanisms [ch12](#)
- Making reactions selective using acids and bases [ch23](#)
- More details on acid and base catalysis [ch39](#)

Organic compounds are more soluble in water as ions

Most organic compounds are insoluble in water. But sometimes it's necessary to make them dissolve, perhaps by converting them to anions or cations. Water can solvate both cations and anions, unlike some of the solvents you will meet later. A good way of dissolving an organic acid is to put it in basic solution: the base deprotonates the acid to give an anion. A simple example is aspirin: whilst the acid itself is not very soluble in water, the sodium salt is much more soluble. The sodium salt forms with the weak base, sodium hydrogencarbonate.

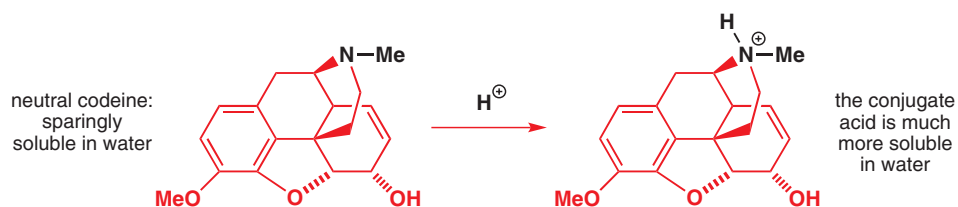


The sodium or calcium salt of 'normal' aspirin is sold as 'soluble aspirin'. But when the pH of a solution of aspirin's sodium salt is lowered, the amount of the 'normal' acidic form present increases and the solubility decreases. In the acidic environment of the stomach (around pH 1–2), soluble aspirin will be converted back to the normal acidic form and precipitate out of solution.

Water is special for many reasons, and it falls into a class of solvents we call *polar protic* solvents. We will discuss other solvents in this class, as well as *polar aprotic* solvents (such as acetone and DMF) and *non-polar* solvents (such as toluene and hexane) in Chapter 12.

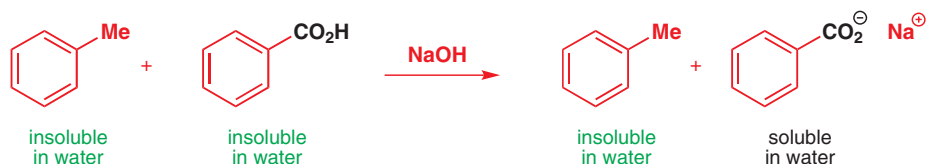


In the same way, organic bases such as amines can be dissolved by *lowering* the pH. Codeine (7,8-didehydro-4,5-epoxy-3-methoxy-17-methylmorphinan-6-ol) is a commonly used pain-killer. Codeine itself is not very soluble in water but it does contain a basic nitrogen atom that can be protonated to give a more soluble salt. It is usually encountered as a phosphate salt. The structure is complex, but that doesn't matter.

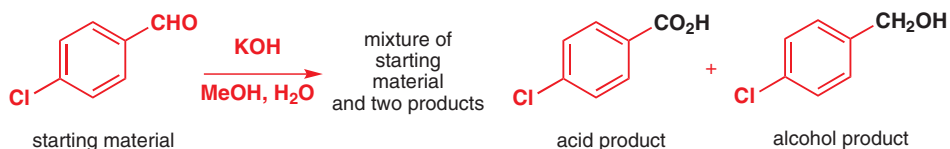


Charged compounds can be separated by acid–base extraction

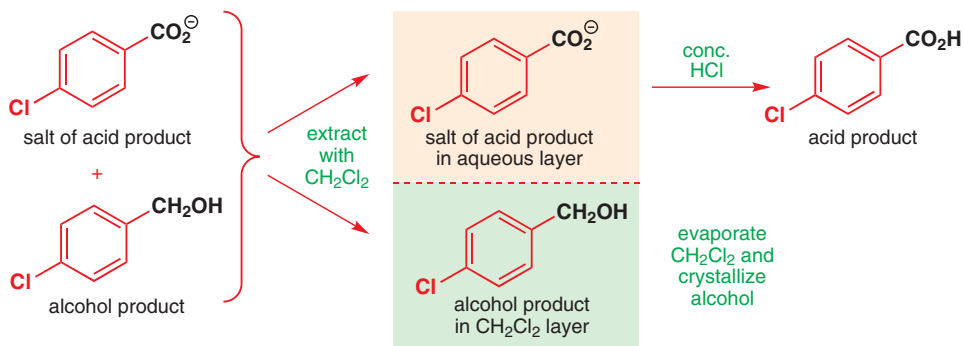
Adjusting the pH of a solution often provides an easy way to separate compounds. Separating a mixture of benzoic acid ($PhCO_2H$) and toluene ($PhMe$) is easy: dissolve the mixture in CH_2Cl_2 , add aqueous NaOH, shake the mixture of solutions, and separate the layers. The CH_2Cl_2 layer contains all the toluene. The aqueous layer contains the sodium salt of benzoic acid. Addition of HCl to the aqueous layer precipitates the insoluble benzoic acid.



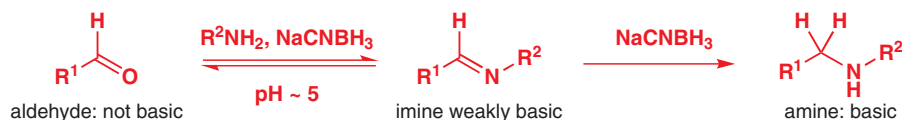
A more realistic separation is given in a modern practical book after a Cannizzaro reaction. You will meet this reaction in Chapters 26 and 39 but all you need to know now is that there are two products, formed in roughly equal quantities. Separation of these from starting material and solvent, as well as from each other, makes this a useful reaction.



The products under the basic reaction conditions are the *salt* of the acid (soluble in water) and the alcohol (not soluble in water). Extraction with dichloromethane removes the alcohol and leaves the salt in the aqueous layer along with solvent methanol and residual KOH. Rotary evaporation of the CH_2Cl_2 layer gives crystalline alcohol and acidification of the aqueous layer precipitates the neutral acid.



In the same way, any basic compounds dissolved in an organic layer can be extracted by washing the layer with dilute aqueous acid and recovered by raising the pH, which will precipitate out the less soluble neutral compound. A general way to make amines is by ‘reductive amination.’ Ignore the details of this reaction for now (we come back to them in Chapter 11) but consider how the amine might be separated from starting material, by-products, and solvent.



As the reaction mixture is weakly acidic, the amine will be protonated and will be soluble in water. The starting material and intermediate (of which very little is present anyway) are soluble in organic solvents. Extracting the aqueous layer and neutralizing with NaOH gives the amine.

Whenever you do any extractions or washes in practical experiments, just stop and ask yourself: ‘What is happening here? In which layer is my compound and why?’ You will then be less likely to throw away the wrong layer (and your precious compound)!

Acids, bases, and pK_a

If we are going to make use of the acid–base properties of compounds as we have just described, we are going to need a way of measuring *how acidic* or *how basic* they are. Raising the pH leads to deprotonation of aspirin and lowering the pH leads to protonation of codeine, but *how far* do we have to raise or lower the pH to do this? The measure of acidity or basicity we need is called pK_a . The value of pK_a tells us how acidic (or not) a given hydrogen atom in a compound is. Knowing about pK_a tells us, for example, that the amine product from the reaction just above will be protonated at weakly acidic pH 5, or that only a weak base (sodium hydrogen carbonate) is needed to deprotonate a carboxylic acid such as aspirin. It is also useful because many reactions proceed through protonation or deprotonation of one of the reactants (you met some examples in Chapter 6), and it is obviously useful to know what strength acid or base is needed. It would be futile to use too weak a base to deprotonate a compound but, equally, using a very strong base where a weak one would do risks the result of cracking open a walnut with a sledge hammer.

The aim of this chapter is to help you to understand *why* a given compound has the pK_a that it does. Once you understand the trends involved, you should have a good feel for the pK_a values of commonly encountered compounds and also be able to predict roughly the values for unfamiliar compounds.

Benzoic acid preserves soft drinks

Benzoic acid is used as a preservative in foods and soft drinks (E210). Like acetic acid, it is only the acid form that is effective as a bactericide. Consequently, benzoic acid can be used as a preservative only in foodstuffs with a relatively low pH, ideally less than its pK_a of 4.2. This isn’t usually a problem: soft drinks, for example, typically have a pH of 2–3. Benzoic acid is often added as the sodium salt (E211), perhaps because this can be added to the recipe as a concentrated solution in water. At the low pH in the final drink, most of the salt will be protonated to give benzoic acid proper, which presumably remains in solution because it is so dilute.

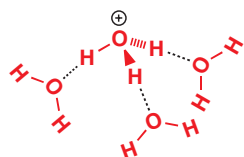
Acidity

Let’s start with two simple, and probably familiar, definitions:

- An acid is a species having a tendency to lose a proton.
- A base is a species having a tendency to accept a proton.

'The proton is a unique chemical species, being a bare nucleus. As a consequence it has no independent existence in the condensed state and is invariably found bound by a pair of electrons to another atom.'

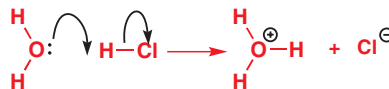
Ross Stewart, *The Proton: Applications to Organic Chemistry*, Academic Press, Orlando, 1985, p. 1.



a structure for a solvated hydronium ion in water: the dashed bonds represent hydrogen bonds

An isolated proton is extremely reactive—formation of H_3O^+ in water

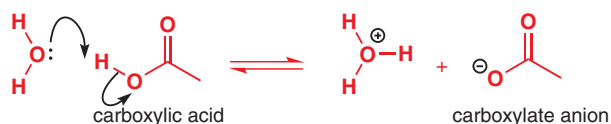
Gaseous HCl is not an acid at all—it shows no tendency to dissociate into H^+ and Cl^- as the H–Cl bond is strong. But hydrochloric acid—that is, a solution of HCl in water—is a strong acid. The difference is that an isolated proton H^+ is too unstable to be encountered under normal conditions, but in water the hydrogen of HCl is transferred to a water molecule and not released as a free species.



The chloride anion is the same in both cases: the only difference is that a very unstable naked proton would have to be the other product in the gas phase but a much more stable H_3O^+ cation would be formed in water. In fact it's even better than that, as other molecules of water cluster round ('solvate') the H_3O^+ cation, stabilizing it with a network of hydrogen bonds.

That is why HCl is an acid in water. But how strong an acid is it? This is where chloride plays a role: hydrochloric acid is a strong acid because chloride ion is a stable anion. The sea is full of it! Water is needed to reveal the acidic quality of HCl, and acidity is determined in water as the standard solvent. If we measure acidity in water, what we are really measuring is how much our acid transfers a proton to a water molecule.

HCl transfers its proton almost completely to water, and is a strong acid. But the transfer of protons to water from carboxylic acids is only partial. That is why carboxylic acids are weak acids. Unlike the reaction of HCl with water, the reaction below is an equilibrium.



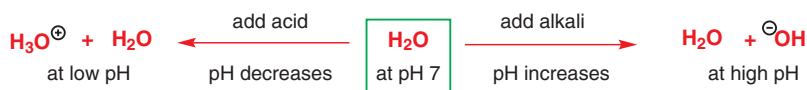
The pH scale and pK_a

The amount of H_3O^+ in any solution in water is described using the pH scale. pH is simply a measure of the concentration of H_3O^+ on a logarithmic scale, and it is characteristic of any aqueous acid—it depends not only on what the acid is (hydrochloric, acetic, etc.) but also on how concentrated the acid is.

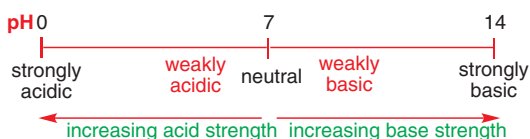
● pH is the negative logarithm of the H_3O^+ concentration.

$$pH = -\log[H_3O^+]$$

You will already know that neutrality is pH 7 and that below pH 7 water is increasingly acidic while above pH 7 it is increasingly basic. At higher pH, there is little H_3O^+ in the solution and more hydroxide ion, but at lower pH there is more H_3O^+ and little hydroxide.



The reason that higher pH means less H_3O^+ is because the arbitrary definition of pH is the *negative* logarithm (to the base 10) of the H_3O^+ concentration. To summarize in a diagram:



■ We will explain later why this scale seems to stop at pH 0 and 14—in fact these numbers are approximate, but easy to remember.

pH is used to measure the acidity of aqueous solutions, but what about the inherent tendency of an acidic compound to give up H^+ to water and form these acid solutions? A good way of measuring this tendency is to find the pH at which a solution contains exactly the same amount of the protonated, acidic form and its deprotonated, basic form. This number, which is characteristic of any acid, is known as the pK_a . In the example just above, this would be the pH where the amount of the carboxylic acid is matched by the amount of its carboxylate salt—which happens to be at about pH 5: the pK_a of acetic acid is 4.76.

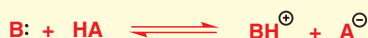
We'll come back to a more formal definition of pK_a later, but first we need to look more closely at this pair of species—the protonated acid and its deprotonated, basic partner.

Every acid has a conjugate base

Looking back at the equilibrium set up when acetic acid dissolves in water, but drawing the mechanism of the back reaction, we see acetate ion acting as a base and H_3O^+ acting as an acid. In all equilibria involving just proton transfer a species acting as a base on one side acts as an acid on the other. We describe H_3O^+ as the *conjugate acid* of water and water as the *conjugate base* of H_3O^+ . In the same way, acetic acid is the conjugate acid of acetate ion and acetate ion is the conjugate base of acetic acid.



● For any acid and any base:

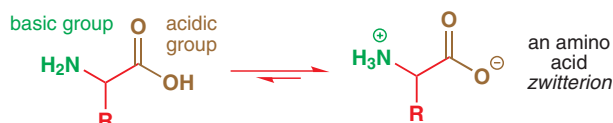


HA is an acid and A^- is its conjugate base and B is a base and BH^+ is its conjugate acid. That is, every acid has a conjugate base associated with it and every base has a conjugate acid associated with it.

Water doesn't have to be one of the participants—if we replace water in the reaction we have been discussing with ammonia, we now have ammonia as the conjugate base of NH_4^+ (the ammonium cation) and the ammonium cation as the conjugate acid of ammonia. What is different is the position of equilibrium: ammonia is more basic than water and now the equilibrium will be well over to the right. As you will see, pK_a will help us assess where equilibria like these lie.



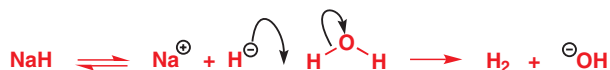
The amino acids you met in Chapter 2 have carboxylic acid and amine functional groups within the same molecule. When dissolved in water, they transfer a proton from the CO_2H group to the NH_2 group and form a *zwitterion*. This German term describes a double ion having positive and negative charges in the same molecule.



Water can behave as an acid or as a base

So far we have seen water acting as a (very weak) base to form H_3O^+ . If we added a strong base, such as sodium hydride, to water, the base would deprotonate the water to give hydroxide ion,

HO^- , and here the water would be acting as an acid. It's amusing to notice that hydrogen gas is the conjugate acid of hydride ion, but more important to note that hydroxide ion is the conjugate base of water.



Water is a weak acid and a weak base so we need a strong acid like HCl to give much H_3O^+ , and a strong base, like hydride ion, to give much hydroxide ion.

The ionization of water

The concentration of H_3O^+ ions in water is very low indeed at $10^{-7} \text{ mol dm}^{-3}$. Pure water at 25°C therefore has a pH of 7.00. Hydronium ions in pure water can arise only from water protonating (and deprotonating) itself. One molecule of water acts as a base, deprotonating another that acts as an acid. For every H_3O^+ ion formed, a hydroxide ion must also be formed, so that in pure water at pH 7 the concentrations of H_3O^+ and hydroxide ions must be equal: $[\text{H}_3\text{O}^+] = [\text{OH}^-] = 10^{-7} \text{ mol dm}^{-3}$.

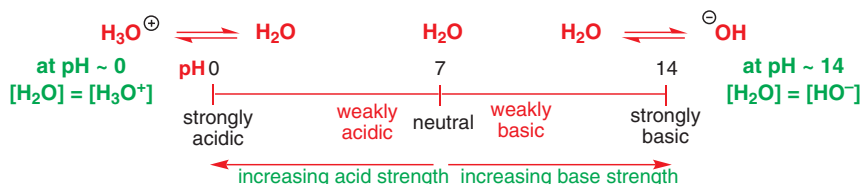


■ Water is still safe to drink because the concentrations of hydronium and hydroxide ions are very small ($10^{-7} \text{ mol dm}^{-3}$ corresponds to about 2 parts per billion). This very low concentration means that there are not enough free hydronium or hydroxide ions in water to do any harm when you drink it, but neither are there enough to provide acid or base catalysts for reactions which need them.

■ The figures 0 and 14 are approximate—there is a simple reason why this is so, which we will explain shortly. But you see now why we end the scale at these points—below 0 and above 14 there is little scope for varying the concentration of H_3O^+ .

The product of these two concentrations is known as the *ionization constant* (or as the *ionic product*) of water, K_w , with a value of $10^{-14} \text{ mol}^2 \text{ dm}^{-6}$ (at 25°C). This is a constant in aqueous solutions, so if we know the hydronium ion concentration (which we can get by measuring the pH), we also know the hydroxide concentration since the product of the two concentrations always equals 10^{-14} .

So, roughly at what pH does water become mostly H_3O^+ ions and at what pH mostly hydroxide ions? We can now add two additional pieces of information to the approximate chart we gave you before. At pH 7, water is almost entirely H_2O . At about pH 0, the concentrations of water and H_3O^+ ions are about the same and at about pH 14, the concentrations of hydroxide ions and water are about the same.



Acids as preservatives

Acetic acid is used as a preservative in many foods, for example pickles, mayonnaise, bread, and fish products, because it prevents bacteria and fungi growing. However, its fungicidal nature is not due to any lowering of the pH of the foodstuff. In fact, it is the undissociated acid that acts as a bactericide and a fungicide in concentrations as low as 0.1–0.3%. Besides, such a low concentration has little effect on the pH of the foodstuff anyway.

Although acetic acid can be added directly to a foodstuff (disguised as E260), it is more common to add vinegar, which contains between 10 and 15% acetic acid. This makes the product more 'natural' since it avoids the nasty 'E numbers'. Actually, vinegar has also replaced other acids used as preservatives, such as propionic (propanoic) acid (E280) and its salts (E281, E282, and E283).

The definition of pK_a

When we introduced you to pK_a on p. 167, we said it is the pH at which an acid and its conjugate base are present in equal concentrations. We can now be more precise about the definition

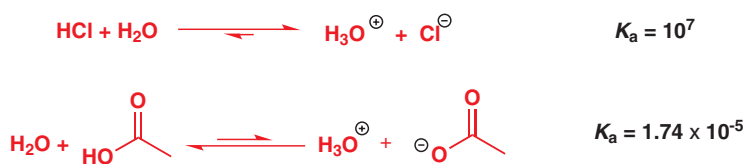
of pK_a . pK_a is the log (to the base ten) of the equilibrium constant for the dissociation of the acid. For an acid HA this is:



How concentrated is water? One mole of pure water has a mass of 18 g and occupies 18 cm³. So, in 1 dm³, there are 1000/18 = 55.56 mol. Water is a 55.56 mol dm⁻³ solution of water...in water.

The concentration of water is ignored in the definition because it is also constant (at 25 °C). Because of the minus sign in the definition (it's there too in the definition of pH) the lower the pK_a the larger the equilibrium constant and the stronger the acid. You may find the way we introduced pK_a more helpful as a concept for visualizing pK_a : any acid is half dissociated in a solution whose pH matches the acid's pK_a . At a pH above the pK_a the acid exists largely as its conjugate base (A^-) but at a pH below the pK_a the acid largely exists as HA.

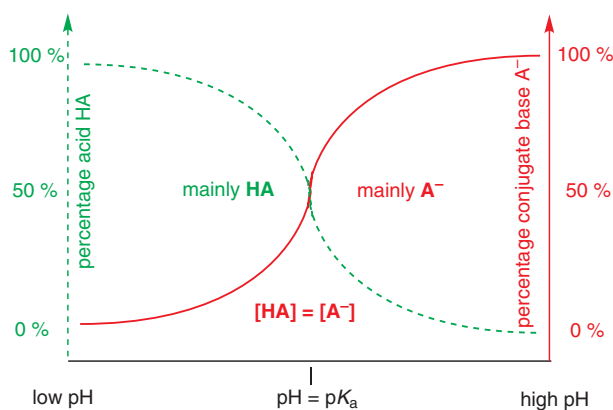
With pK_a we can put figures to the relative strengths of hydrochloric and acetic acid we introduced earlier. HCl is a much stronger acid than acetic acid: the pK_a of HCl is around -7 compared to 4.76 for acetic acid. This tells us that in solution K_a for hydrogen chloride is 10⁷ mol dm⁻³. This is an enormous number: only one molecule in 10,000,000 is not dissociated, so it is essentially fully dissociated. But K_a for acetic acid is only 10^{-4.76} = 1.74 × 10⁻⁵ mol dm⁻³ so it is hardly dissociated at all: only a few molecules in every million of acetic acid are present as the acetate ion.



What about the pK_a of water? You know the figures already: K_a for water is $[H_3O^+] \times [HO^-]/[H_2O] = 10^{-14}/55.5$. So $pK_a = -\log[10^{-14}/55.5] = 15.7$. Now you see why water isn't really quite half dissociated at pH 14—the concentration of water in the equation means that the two ends of the scale on p. 168 are not at 0 and 14, but at -1.7 and 15.7.

A graphical description of the pK_a of acids and bases

For both cases, adjusting the pH alters the proportions of the acid form and of the conjugate base. The graph plots the concentration of the free acid AH (green curve) and the ionized conjugate base A^- (red curve) as percentages of the total concentration as the pH is varied. At low pH the compound exists entirely as AH and at high pH entirely as A^- . At the pK_a the concentration of each species, AH and A^- , is the same. At pHs near the pK_a the compound exists as a mixture of the two forms.



Now we have established why you need to understand acids and bases, we must move on to consider why some acids are stronger than other acids and some bases stronger than other bases. To do this we must be able to estimate the pK_a of common classes of organic compounds.

You do not need to learn exact figures for pK_a values, but you will certainly need to develop a feel for approximate values—we will guide you towards which figures are worth learning and which you can leave to be looked up when you need them.

An acid's pK_a depends on the stability of its conjugate base

The stronger the acid, the easier it is to ionize, which means that it must have a stable conjugate base. Conversely, a weak acid is reluctant to ionize because it has an *unstable* conjugate base. The other side of this coin is that unstable anions A^- make strong bases and their conjugate acids AH are weak acids.

● Acid and conjugate base strength

The stronger the acid HA , the weaker its conjugate base A^- .

The stronger the base A^- , the weaker its conjugate acid AH .

For example, hydrogen iodide has a very low pK_a , about -10 . This means that HI is a strong enough acid to protonate almost anything. Its conjugate base, iodide ion, is therefore not basic at all—it will not deprotonate anything. A very powerful base is methyllithium, $MeLi$. Although it is actually a covalent compound, as we discuss in Chapter 9, for the purpose of the discussion here you can think of $MeLi$ as CH_3Li^+ . CH_3^- can accept a proton to become neutral methane, CH_4 . Methane is therefore the conjugate acid. Clearly, methane isn't at all acidic—its pK_a is estimated to be 48. The table below gives a few inorganic compounds and their approximate pK_a values.

The approximate pK_a values of some inorganic compounds

Acid	pK_a	Conjugate base	Acid	pK_a	Conjugate base	Acid	pK_a	Conjugate base
H_2SO_4	-3	HSO_4^-	H_3O^+	-1.7	H_2O	NH_4^+	9.2	NH_3
HCl	-7	Cl^-	H_2O	15.7	HO^-	NH_3	33	NH_2^-
HI	-10	I^-	H_2S	7.0	HS^-			

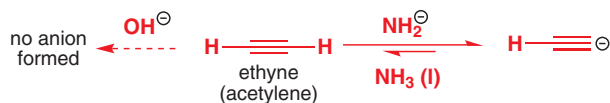
Notice that the lower down the periodic table we go, the stronger the acid. Notice also that oxygen acids are stronger than nitrogen acids. We have also put down more exact pK_a values for water but you need remember only the approximate values of 0 and 14. Over the next few pages we shall be considering the reasons for these differences in acid strength but we are first going to consider the simple consequences of mixing acids or bases of different strengths. Notice the vast range covered by pK_a values: from around -10 for HI to nearly 50 for methane. This corresponds to a difference of 10^{60} in the equilibrium constant.

The choice of solvent limits the pK_a range we can use

In water, we can measure the pK_a of an acid only if the acid does not completely protonate water to give H_3O^+ or completely deprotonate it to give HO^- . We are restricted roughly to pH -1.7 to 15.7 , beyond which water is more than 50% protonated or deprotonated. The strength of acids or bases we can use in any solvent is limited by the acidity and basicity of the solvent itself. Think of it this way: say you want to remove the proton from a compound with a high pK_a , say 25–30. It would be impossible to do this in water since the strongest base we can use is hydroxide. If you add a base stronger than hydroxide, it won't deprotonate your compound, it will just deprotonate water and make hydroxide anyway. Likewise, acids stronger than H_3O^+ can't exist in water: they just protonate water completely to make H_3O^+ . If you do need a stronger base than OH^- (or a stronger acid than H_3O^+ , but this is rarer) you must use a different solvent.

Let's take acetylene as an example. Acetylene (ethyne) has pK_a 25. This is remarkably low for a hydrocarbon (see below for why) but, even so, hydroxide (the strongest base we could have in aqueous solution, pK_a 15.7) would establish an equilibrium where only 1 in $10^{9.3}$ ($10^{15.7}/10^{25}$), or about 1 in 2 billion, ethyne molecules are deprotonated. We can't use a stronger base than hydroxide, since, no matter what strong base we dissolve in water, we will only at best get hydroxide ions. So, in order to deprotonate ethyne to any appreciable extent, we must use a different solvent—one that does not have a pK_a less than 25.

Conditions often used to do this reaction are sodium amide (NaNH_2) in liquid ammonia. Using the pK_a values of NH_3 (ca. 33) and ethyne (25) we would estimate an equilibrium constant for this reaction of 10^8 ($10^{-25}/10^{-33}$)—well over to the right. Amide ions can be used to deprotonate alkynes.



Since we have an upper and a lower limit on the strength of an acid or base that we can use in water, this poses a bit of a problem: how do we know that the pK_a for HCl is more negative than that of H_2SO_4 if both completely protonate water? How do we know that the pK_a of methane is greater than that of ethyne since both the conjugate bases fully deprotonate water? The answer is that we can't simply measure the equilibrium for the reaction in water—we can do this only for pK_a values that fall between the pK_a values of water itself. Outside this range, pK_a values are determined in other solvents and the results are extrapolated to give a value for what the pK_a in water might be.

Constructing a pK_a scale

We now want to look at ways to rationalize, and estimate, the different pK_a values for different compounds—we wouldn't want to have to memorize all the values. You will need to get a feel for the pK_a values of different compounds and if you know what factors affect them it will make it much easier to predict an approximate pK_a value, or at least understand why a given compound has the pK_a value that it does.



A number of factors affect the strength of an acid AH . These include:

1. The intrinsic stability of the conjugate base, anion $\text{A}^\ominus$. Stability can arise by having the negative charge on an electronegative atom or by spreading the charge over several atoms (delocalization) groups. Either way, the more stable the conjugate base, the stronger the acid HA .
2. Bond strength A-H . Clearly, the easier it is to break this bond, the stronger the acid.
3. The solvent. The better the solvent is at stabilizing the ions formed, the easier it is for the reaction to occur.

● Acid strength

The most important factor in the strength of an acid is the stability of the conjugate base—the more stable the conjugate base, the stronger the acid.

An important factor in the stability of the conjugate base is which element the negative charge is on—the more electronegative the element, the more stable the conjugate base.

The negative charge on an electronegative element stabilizes the conjugate base

The pK_a values for the 'hydrides' of the first row elements CH_4 , NH_3 , H_2O , and HF are about 48, 33, 16, and 3, respectively. This trend is due to the increasing electronegativities across the period: $\text{F}^\ominus$ is much more stable than $\text{CH}_3^\ominus$, because fluorine is much more electronegative than carbon.

Acid	Conjugate base	pK_a
methane CH_4	$\text{CH}_3^\ominus$	~48
ammonia NH_3	amide ion $\text{NH}_2^\ominus$	~33
water H_2O	hydroxide ion $\text{HO}^\ominus$	~16
HF	fluoride ion $\text{F}^\ominus$	3

■ Because the pK_a values for very strong acids and bases are so hard to determine, you will find that they often differ in different texts—sometimes the values are no better than good guesses! However, while the absolute values may differ, the relative values (which is the important thing because we need only a rough guide) are usually consistent.

Weak A–H bonds make stronger acids

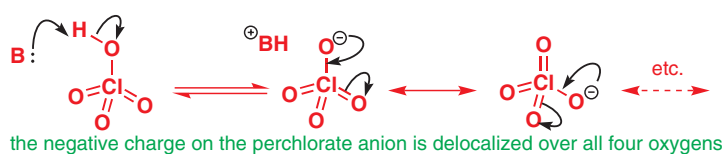
However, on descending group VII (group 17), the pK_a values for HF, HCl, HBr, and HI decrease: 3, –7, –9, and –10. Since the electronegativities decrease on descending the group we might expect an increase in pK_a . The decrease is due to the weakening bond strengths on descending the group and to some extent the way in which the charge can be spread over the increasingly large anions.

Acid	Conjugate base	pK_a
HF	fluoride ion F^-	3
HCl	chloride ion Cl^-	–7
HBr	bromide ion Br^-	–9
HI	iodide ion I^-	–10

Delocalization of the negative charge stabilizes the conjugate base

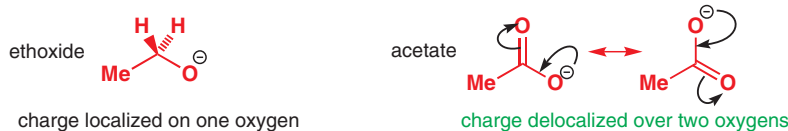
The acids $HClO$, $HClO_2$, $HClO_3$, and $HClO_4$ have pK_a values 7.5, 2, –1, and about –10, respectively. In each case the acidic proton is on an oxygen attached to chlorine, that is, *we are removing a proton from the same environment in each case*. Why then is perchloric acid, $HClO_4$, some 17 orders of magnitude stronger in acidity than hypochlorous acid, $HClO$? Once the proton is removed, we end up with a negative charge on oxygen. For hypochlorous acid, this is localized on the one oxygen. With each successive oxygen, the charge can be more delocalized, and this makes the anion more stable. For example, with perchloric acid, the negative charge can be delocalized over all four oxygen atoms.

Acid	Conjugate base	pK_a
hypochlorous acid $HO-Cl$	ClO^-	7.5
chlorous acid $HO-ClO$	ClO_2^-	2
chloric acid $HO-ClO_2$	ClO_3^-	–1
perchloric acid $HO-ClO_3$	ClO_4^-	–10

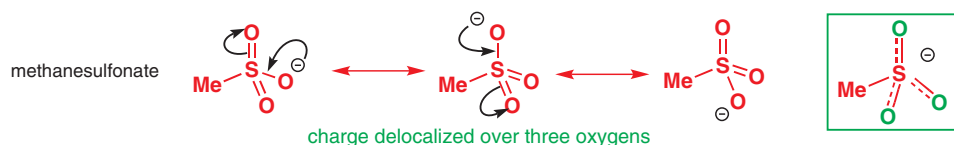


That the charge is spread out over all the oxygen atoms equally is shown by electron diffraction studies: whereas perchloric acid has two types of $Cl-O$ bond, one 163.5 pm and the other three 140.8 pm long, in the perchlorate anion all $Cl-O$ bond lengths are the same, 144 pm, and all $O-Cl-O$ bond angles are 109.5° . *Just to remind you:* these delocalization arrows do not indicate that the charge is actually moving from atom to atom. We discussed this in Chapter 7. These structures simply show that the charge is spread out in the molecular orbitals and mainly concentrated on the oxygen atoms.

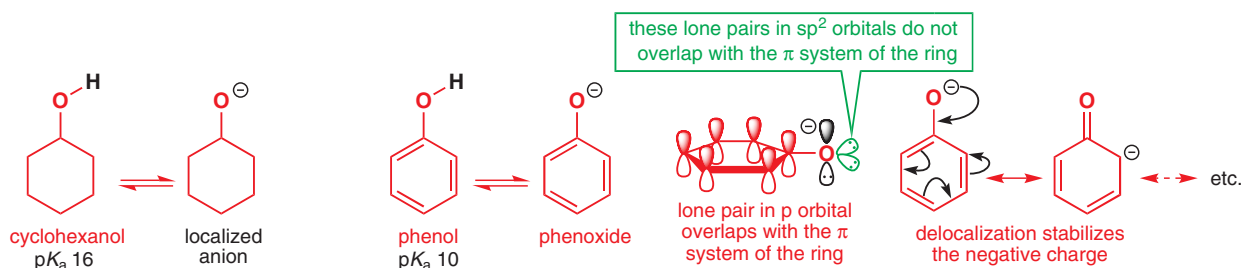
Looking at some organic acids, we might expect alcohols to have a pK_a not far from that of water, and for ethanol that is correct (pK_a 15.9). If we allow the charge in the conjugate base to be delocalized over two oxygen atoms, as in acetate, acetic acid is indeed a much stronger acid (pK_a 4.8). The difference is huge: the conjugation makes acetic acid about 10^{10} times stronger.



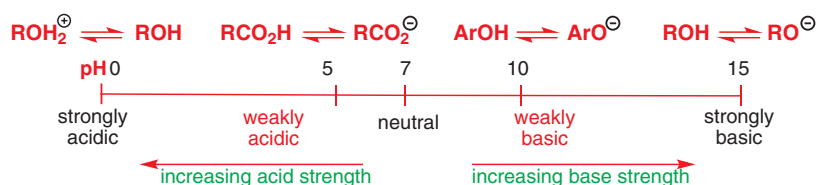
It is even possible to have a negative charge of an organic acid delocalized over *three* atoms—as in the anions of the sulfonic acids. Methanesulfonic acid has a pK_a of -1.9 .



Even delocalization into a hydrocarbon part of the molecule increases acid strength. In phenol, PhOH , the OH group is directly attached to a benzene ring. On deprotonation, the negative charge can again be delocalized, not onto other oxygen atoms but into the aromatic ring itself. The effect of this is to stabilize the phenoxide anion relative to the conjugate base of cyclohexanol, where no delocalization is possible, and this is reflected in the pK_a values of the two compounds: 10 for phenol but 16 for cyclohexanol.



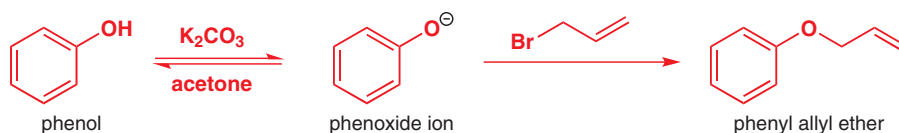
So now we can expand our chart of acid and base strengths to include the important classes of alcohols, phenols, and carboxylic acids. They conveniently, and memorably, have pK_a values of about 0 for the protonation of alcohols, about 5 for the deprotonation of carboxylic acids, about 10 for the deprotonation of phenols, and about 15 for the deprotonation of alcohols. The equilibria above each pK_a shows that at approximately that pH, the two species each form 50% of the mixture. You can see that carboxylic acids are weak acids, alkoxide ions (RO^-) are strong bases, and that it will need a strong acid to protonate an alcohol.



■ equilibrium arrow: $\rightleftharpoons$
 delocalization arrow: $\longleftrightarrow$
 Reminder: the equilibrium arrows mean two interconverting compounds. The double-headed arrow means two ways of drawing a conjugated structure.

■ It is worthwhile learning these approximate values.

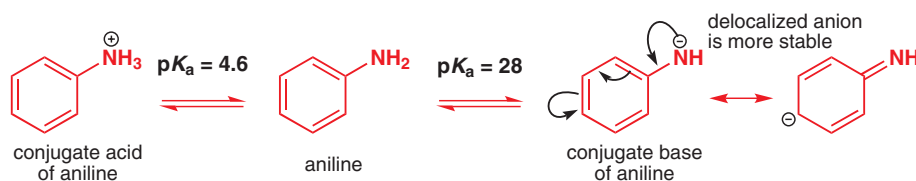
If we need to make the anion of a phenol, a base such as NaOH will be good enough, but if we want to make an anion from an alcohol, we need a stronger base. Vogel (p. 986) suggests potassium carbonate (K_2CO_3) is strong enough to make an ether from phenol. The base strength of carbonate anion is about the same as that of phenoxide ion (PhO^-) so the two will be in equilibrium but enough phenoxide ion will be present for the reaction.



On the other hand, if we want to make the OH group into a good leaving group, we need to protonate it and a very strong acid will be needed. Sulfuric acid is used to make ethers from alcohols. Protonation of the OH groups leads to loss of water and formation of a cation. This reacts with more alcohol to give the ether. There is another example of this reaction in Chapter 5.

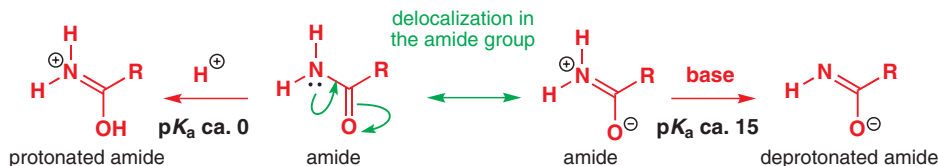
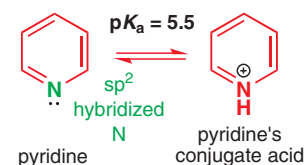
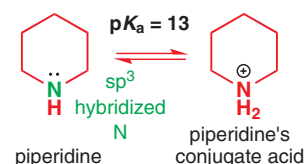
■ As you will discover in Chapters 10 and 15, a *leaving group* is simply a functional group that will leave the molecule, taking with it the pair of electrons that formed the bond. Leaving groups may be anions, such as bromide Br^- , or protonated groups such as the protonated alcohol in this example, which leaves as water.

But for the same reason, aniline is also more acidic than ammonia (pK_a 33) and has a genuine pK_a in which one of the protons on nitrogen is lost. So we can say correctly that ‘the pK_a of aniline is about 28.’ Just be careful to check which pK_a is meant in such compounds. The full picture is:



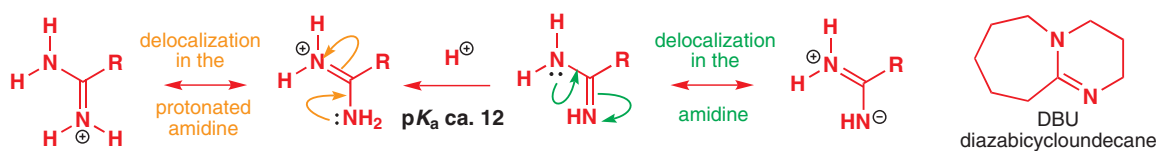
The pK_a associated for protonation of piperidine, a typical secondary amine, is about 13. The equivalent pK_a for protonation of pyridine—a compound with a similar heterocyclic structure, but with its lone pair in an sp^2 rather than an sp^3 orbital, is only 5.5: pyridine is a weaker base than piperidine (its conjugate acid is a stronger acid). Nitriles, whose lone pair is sp hybridized, are not basic at all. Lone pairs with more p character (sp^3 orbitals are $3/4$ p , while sp orbitals are $1/2$ p) are higher in energy—they spend more time further from the nucleus—and are therefore more basic.

Amides are very different because of the delocalization of the lone pair into the carbonyl group. This makes amides more acidic but less basic and protonation occurs on oxygen rather than nitrogen. Amides have pK_a values of around 15 when they act as acids, making them some 10^{10} times more acidic than amines. The pK_a of protonated amides is around 0, making them some 10^{10} times weaker as bases.

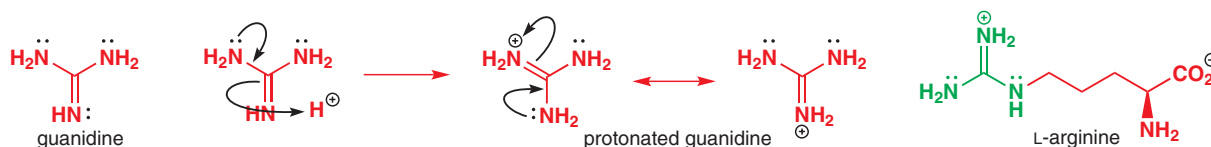


→ Delocalization in amides was discussed on p. 155.

If we replace the carbonyl oxygen atom in an amide by nitrogen we get an amidine. Amidines are conjugated, like amides, but unlike amides they are *stronger* bases than amines, by about 2–3 pK_a units, because the two nitrogens work together to donate electron density onto each other. The bicyclic amidine DBU is often used as a strong organic base (see Chapter 17).



But the champions are the guanidines, with three nitrogens all donating lone pair electrons at once. A guanidine group (shown in green) makes arginine the most basic of the amino acids.

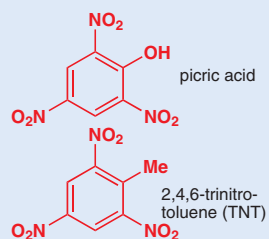


Substituents affect the pK_a

Substituents that are conjugated with the site of proton gain or loss, and even substituents that are electronegative but not conjugating, can have significant effects on pK_a values. Phenol has pK_a 10 but phenols with anions stabilized by extra conjugation can have much lower pK_a s.

Picric acid is a very acidic phenol

2,4,6-Trinitrophenol's more common name, picric acid, reflects the strong acidity of this compound (pK_a 0.7 compared to phenol's 10.0). Picric acid used to be used in the dyeing industry but is little used now because it is also a powerful explosive when dry. (Compare its structure with that of TNT!)

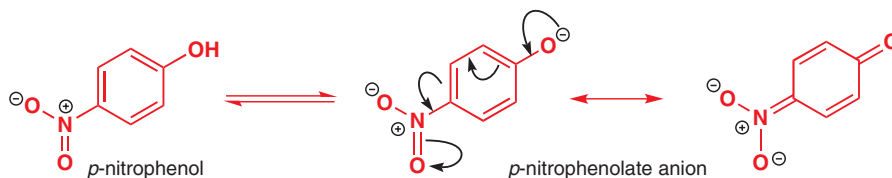


■ If you draw a carboxylate anion you will find that it is impossible to stabilize its negative charge any further by conjugation, other than between the two oxygens.

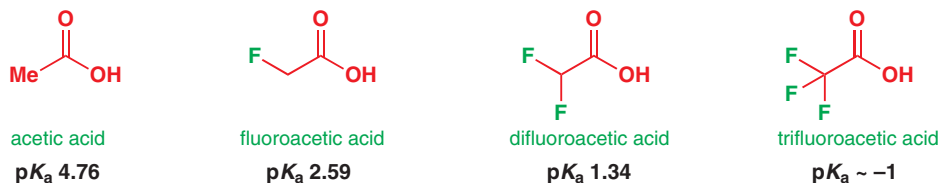


■ If you don't see why this is, think about the shapes of an s and a p orbital: the nucleus sits in the node of a p orbital, but in an s orbital the nucleus is in a region of high electron density. The more s character a negative charge has, the closer the electron density is to the nucleus, and the more stable it is.

One nitro group, as in *p*-nitrophenol, lowers the pK_a to 7.14, nearly a thousand-fold increase in acidity. This is because the negative charge on oxygen is delocalized into the very electron-withdrawing nitro group. By contrast 4-chlorophenol, with only inductive withdrawal in the C–Cl bond, has pK_a 9.38, hardly different from phenol itself.



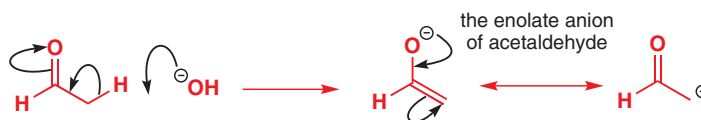
Inductive effects of nearby electronegative atoms can also have marked effects on the pK_a of acids. Adding fluorines to acetic acid reduces the pK_a from about 5 by smallish steps. Trifluoroacetic acid (TFA) is a very strong acid indeed, and is commonly used as a convenient strong acid in organic reactions. Inductive effects occur by polarization of σ bonds when the atom at one end is more electronegative than at the other. Fluorine is much more electronegative than carbon (indeed, F is the most electronegative element of all) so each σ bond is very polarized, making the carbon atom more electropositive and stabilizing the carboxylate anion.



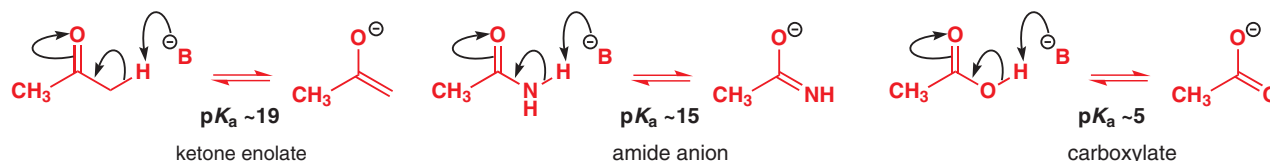
Carbon acids

Hydrocarbons are not acidic. We have already established that methane has a pK_a of about 48 (p. 170 above)—it's essentially impossible to deprotonate. Alkylolithiums are for this reason among the strongest bases available. But some hydrocarbons *can* be deprotonated, the most important example being alkynes—you saw on p. 171 that acetylene has a pK_a of 25 and can be deprotonated by NH_2^- (as well as other strong bases such as BuLi). The difference is one of hybridization—an idea we introduced with the nitrogen bases above. Making the acetylide anion, whose negative charge resides in an sp orbital, is much easier than making a methyl anion, with a negative charge in an sp^3 orbital, because electrons in sp orbitals spend a lot of their time closer to the nucleus than electrons in sp^3 orbitals.

C–H bonds can be even more acidic than those of acetylene if stabilization of the resulting anion is possible by *conjugation*. Conjugation with a carbonyl group has a striking effect. One carbonyl group brings the pK_a down to 13.5 for acetaldehyde so that even hydroxide ion can produce the anion. You will discover in Chapter 20 that we call this the 'enolate anion' and that the charge is mostly on oxygen, although the anion can be drawn as a carbanion.

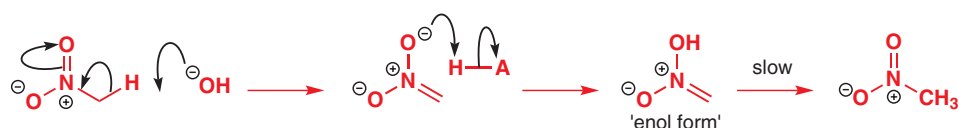


It is interesting to compare the strengths of the carbon, nitrogen, and oxygen acids of similar structure below. The ketone (acetone) is of course least acidic, the amide is more acidic, and the carboxylic acid most acidic. The oxyanion conjugate bases are all delocalized but delocalization onto a second very electronegative oxygen atom is much (~10 pH units) more effective than delocalization onto nitrogen, which is 4 pH units more effective than delocalization onto carbon.



Nevertheless, the effect of conjugation on the carbon acid compared with methane is enormous (~30 pH units) and brings proton removal from carbon within the range of accessible bases.

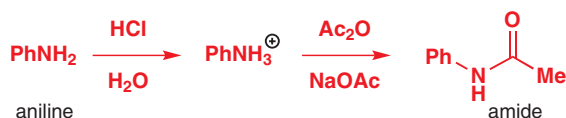
The nitro group is even more effective: nitromethane, with a pK_a of 10, dissolves in aqueous NaOH. The proton is removed from carbon, but the negative charge in the conjugate base is on oxygen. The big difference is that the nitrogen atom has a positive charge throughout. If the anion is protonated in water by some acid (HA) the 'enol' form of nitromethane is the initial product and this slowly turns into nitromethane itself. Whereas proton transfers between electronegative atoms (O, N, etc.) are fast, proton transfers to or from carbon can be slow.



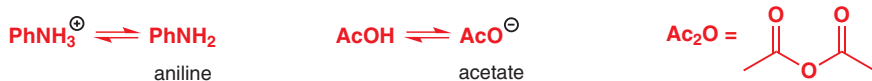
Carbon acids are very important in organic chemistry as they allow us to make carbon-carbon bonds and you will meet many more of them in later chapters of this book.

Why do we need to compare acid strengths of O and N acids?

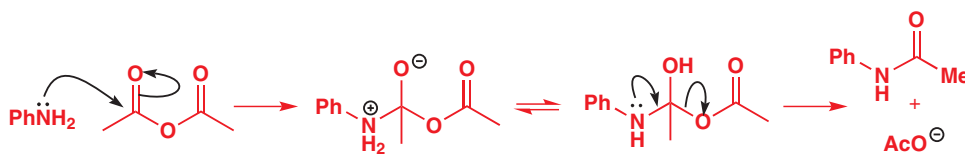
The rates of nucleophilic addition to carbonyl groups that you met in Chapter 6 depend on the basicity of nucleophiles. As nitrogen bases are much stronger than oxygen bases (or, if you prefer, ammonium ions are much weaker acids than H_3O^+), amines are also much better nucleophiles than water or alcohols. This is dramatically illustrated in an amide synthesis from aniline and acetic anhydride in aqueous solution.



Aniline is not very soluble in water but addition of HCl converts it into the soluble cation by protonation at nitrogen. The solution is now warmed and equal amounts of acetic anhydride and aqueous sodium acetate are added. The pK_a of acetic acid is about 5, as is the pK_a of $PhNH_3^+$, so an equilibrium is set up and the solution now contains these species:



The only electrophile is acetic anhydride, with its two electrophilic carbonyl groups. The nucleophiles available are water, aniline, and acetate. Water is there in great abundance and does react with acetic anhydride but can't compete with the other two as they are more basic (by about 10^5). If acetate attacks the anhydride, it simply regenerates acetate. But if aniline attacks, the amide is formed as acetate is released.



The isolation of the product is easy as the amide is insoluble in water and can be filtered off. Environmental considerations suggest that we should not use organic solvents so much and should use water when possible. If we have some idea about pK_a s we can estimate whether water will interfere in a reaction we are planning and decide whether it is a suitable solvent or not. It is even possible to acylate amines with the more reactive acid chlorides in aqueous solution, and we will return in detail to acylation reactions such as these in Chapter 10.

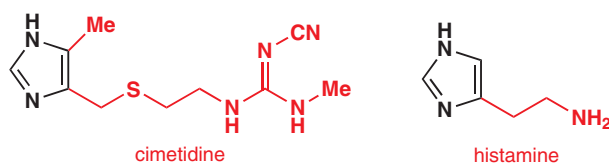
pK_a in action—the development of the drug cimetidine

Histamine is an agonist in the production of gastric acid. It binds to specific sites (receptor sites) in the stomach cells and triggers the production of gastric acid (mainly HCl).

An antagonist works by binding to the receptor but not stimulating acid secretion. It therefore inhibits acid secretion by blocking the receptor sites.

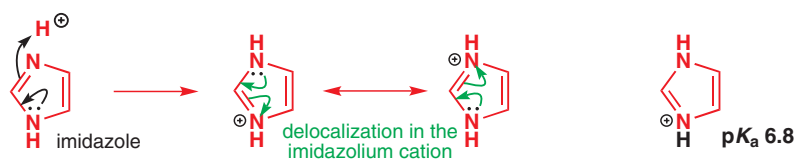
■ When the drug was invented, the company was called Smith, Kline and French (SKF) but after a merger with Beechams the company became SmithKline Beecham (SB). SB and GlaxoWellcome later merged to form GlaxoSmithKline (GSK). Things may have changed further by the time you read this book.

The development of the anti-peptic ulcer drug cimetidine gives a fascinating insight into the important role of pK_a in chemistry. Peptic ulcers are a localized erosion of the mucous membrane, resulting from overproduction of gastric acid in the stomach. One of the compounds that controls the production of the acid is histamine. (Histamine is also responsible for the symptoms of hay fever and allergies.)



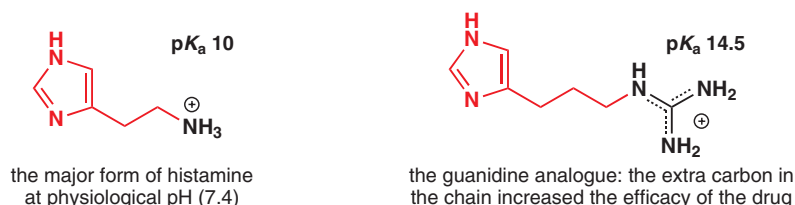
Histamine works by binding into a receptor in the stomach lining and stimulating the production of acid. What the developers of cimetidine at Smith, Kline and French wanted was a drug that would bind to these receptors without activating them and thereby prevent histamine from binding but not stimulate acid secretion itself. Unfortunately, the antihistamine drugs successfully used in the treatment of hay fever did not work—a different histamine receptor was involved.

Notice that cimetidine and histamine both have the same nitrogen-containing ring (shown in black) as part of their structures. This ring is known as an imidazole—imidazole itself is quite a strong base whose protonated form is delocalized as shown below. This is not coincidence—cimetidine's design was centred around the structure of histamine.



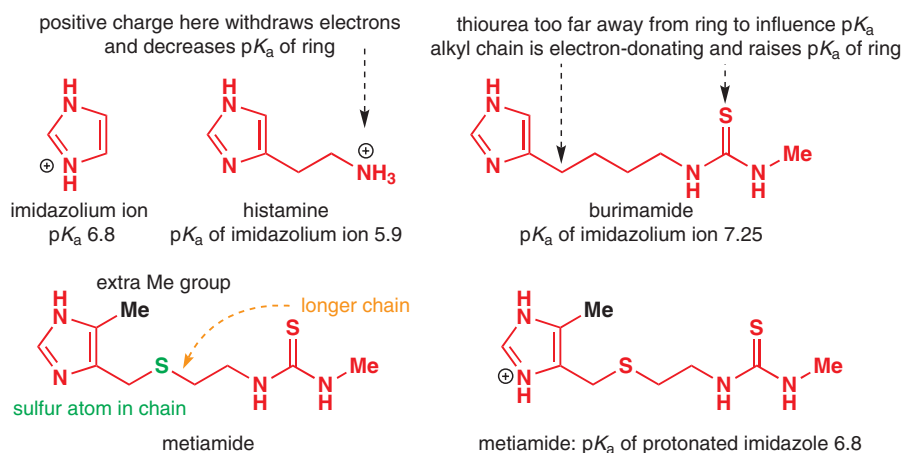
➡ Guanidine was introduced to you on p. 175.

In the body, most histamine exists as a salt, being protonated on the primary amine and the early compounds modelled this. The guanidine analogue was synthesized and tested to see if it had any antagonistic effect (that is, if it could bind in the histamine receptors and prevent histamine binding). It did bind but unfortunately it acted as an *agonist* rather than an *antagonist* and stimulated acid secretion rather than blocking it. Since the guanidine analogue has a pK_a even greater than histamine (about 14.5 compared to about 10), it is effectively all protonated at physiological pH.

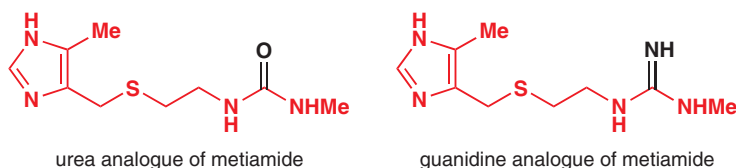


■ Remember that amidines and guanidines, p. 175, are basic but that amides aren't. The thiourea, and indeed a urea, is more like an amide.

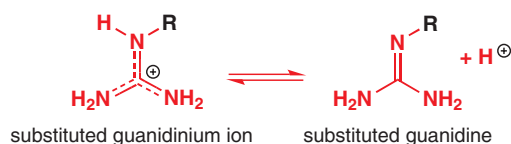
The agonistic behaviour of the drug clearly had to be suppressed. The thought occurred to the chemists that perhaps the positive charge made the compound agonistic, and so a polar but much less basic compound was sought. Eventually, they came up with burimamide. The most important change is the replacement of the $C=NH$ in the guanidine compound by $C=S$. Now instead of a guanidine we have a thiourea, which is much less basic. Other adjustments were to increase the chain length, insert a second sulfur atom on the chain, and add methyl groups to the thiourea and the imidazole ring, to give metiamide with increased efficacy.



The new drug, metiamide, was ten times more effective than burimamide when tested in humans. However, there was an unfortunate side-effect: in some patients the drug caused a decrease in the number of white blood cells, leaving the patient open to infection. This was eventually traced back to the thiourea group. The sulfur had again to be replaced by oxygen, to give a normal urea and, just to see what would happen, by nitrogen to give another guanidine.

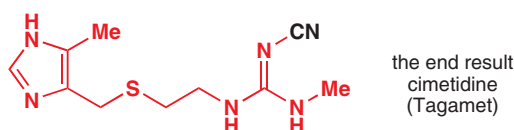


Neither was as effective as metiamide but the important discovery was that the guanidine analogue no longer showed the agonistic effects of the earlier guanidine. Of course, the guanidine would also be protonated so we had the same problem we had earlier—how to decrease the pK_a of the guanidinium ion. A section of this chapter considered the effect of electron-withdrawing groups on pK_a and showed that they make a base less basic. This was the approach now adopted—the introduction of electron-withdrawing groups on to the guanidine to lower its pK_a. The table below shows the pK_as of various substituted guanidinium ions.

pK_as of substituted guanidinium ions

R	H	Ph	CH ₃ CO	NH ₂ CO	MeO	CN	NO ₂
pK _a	14.5	10.8	8.33	7.9	7.5	-0.4	-0.9

Clearly, the cyano and nitro-substituted guanidines would not be protonated at all. These were synthesized and found to be just as effective as metiamide but without the side-effects. Of the two, the cyanoguanidine compound was slightly more effective and this was developed and named 'cimetidine'.



The development of cimetidine by Smith, Kline and French from the very start of the project up to its launch on the market took 13 years. This enormous effort was well rewarded—Tagamet (the trade name of the drug cimetidine) became the best-selling drug in the world and the first to gross more than one billion dollars per annum. Thousands of ulcer patients worldwide no longer had to suffer pain, surgery, or even death. The development of cimetidine followed a rational approach based on physiological and chemical principles and it was for this that one of the scientists involved, Sir James Black, received a share of the 1988 Nobel Prize for Physiology or Medicine. None of this would have been possible without an understanding of pK_a s.

Lewis acids and bases

Johannes Nicolaus Brønsted (1879–1947) was a Danish physical chemist who, simultaneously with Thomas Lowry, introduced the protic theory of acid–base reactions in 1923.

All the acids and bases we have been discussing so far have been protic, or *Brønsted*, acids and bases. In fact, the definition of an acid and a base we gave you on p. 165 is a definition of a Brønsted acid and a Brønsted base. When a carboxylic acid gives a proton to an amine, it is acting as a Brønsted acid while the amine is a Brønsted base. The ammonium ion produced is a Brønsted acid while the carboxylate anion is a Brønsted base.



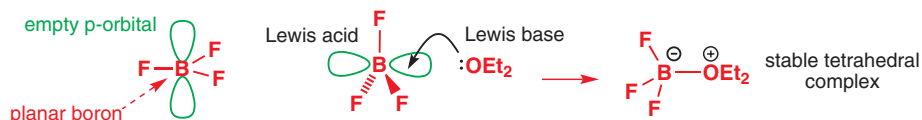
- Brønsted acids donate protons.
- Brønsted bases accept protons.

But there is another important type of acid: the Lewis acid. These acids don't donate protons—indeed they usually have no protons to donate. Instead they accept electrons. It is indeed a more general definition of acids to say that they accept electrons and of bases that they donate electrons. Lewis acids are usually halides of the higher oxidation states of metals, such as BF_3 , AlCl_3 , ZnCl_2 , SbF_5 , and TiCl_4 . By removing electrons from organic compounds, Lewis acids act as important catalysts in important reactions such as the Friedel–Crafts alkylation and acylation of benzene (Chapter 21), the $\text{S}_{\text{N}}1$ substitution reaction (Chapter 15), and the Diels–Alder reaction (Chapter 34).

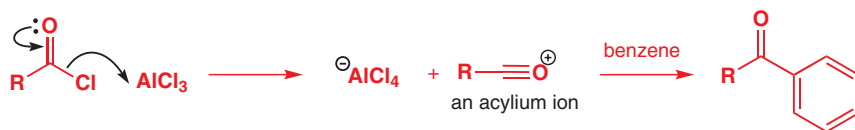
The American chemist Gilbert Lewis (1875–1946) introduced his electronic theory of acid–base interactions in 1924.

- Lewis acids accept electrons.
- Lewis bases donate electrons.

A simple Lewis acid is BF_3 . As you saw in Chapter 5, monomeric boron compounds have three bonds to other atoms and an empty p orbital, making six electrons only in the outer shell. They are therefore not stable and BF_3 is normally used as its 'etherate': a complex with Et_2O . Ether donates a pair of electrons into the empty p orbital of BF_3 and this complex has tetrahedral boron with eight electrons. In this reaction the ether donates electrons (it can be described as a Lewis base) and BF_3 accepts electrons: it is a Lewis acid. No protons are exchanged. The complex is a stable liquid and is the form usually available from suppliers.



Lewis acids often form strong interactions with electronegative atoms such as halides or oxygen. In the Friedel–Crafts acylation, which you will meet in Chapter 21, for example, AlCl_3 removes the chloride ion from an acyl chloride to give a species, the acylium ion, which is reactive enough to combine with benzene.



Lewis acid–base interactions are very common in chemistry and are often rather subtle. You are about to meet, in the next chapter, an important way of making C–C bonds by adding organometallics to carbonyl compounds, and in many of these reactions there is an interaction at some point between a Lewis acidic metal cation and a Lewis basic carbonyl group.

Further reading

The quote at the start of the chapter comes from Ross Stewart, *The Proton: Applications to Organic Chemistry*, Academic Press, Orlando, 1985, p 1.

More detailed information about acid/base extraction can be found in any organic practical book. The details of the Cannizzaro reaction are from J. C. Gilbert and S. F. Martin, *Experimental Organic Chemistry*, Harcourt, Fort Worth, 2002. The reduction of amides to amines comes from B. S. Furniss, A. J. Hannaford, P. W. G. Smith,

and A. R. Tatchell, *Vogel's Textbook of Practical Organic Chemistry*, 5th edn, Longman, Harlow, 1989.

Details about the acylation of amines with anhydrides and acid chlorides are in L. M. Harwood, C. J. Moody, and J. M. Percy, *Experimental Organic Chemistry*, 2nd edn, Blackwell, Oxford, 1999, p 279.

There is more about the discovery of cimetidine in W. Sneader, *Drug Discovery: a History*, Wiley, Chichester, 2005.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>